

Anti-bluff correction “ Option 4 “rootless container survives user@1000 SIGKILL” was WRONG

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Governing anchors: Â§11.4.6 (no-guessing / anti-guessing) Â· Â§11.4.138 (operator-escape + bluff-audit + permanent guard) Â· Â§11.4.194 (exhaustive all-scenario, all-angle code-review, prove every assumption) Â· Â§11.4.226 (evidence-class-at-closure) Â· Â§11.4.201 (guards MUST assert the REAL condition “ a false-POSITIVE architectural claim is a FAIL-bluff equal in severity to a false-negative)

Classification: self-authored Â§11.4.138-class correction. No operator had to catch this “ it was caught by a subsequent Phase-1.5 spike whose captured cgroup / systemctl evidence refuted an earlier architectural claim made in this same host’s incident-response work.

1. What I claimed

While drafting the forced-logout preventive-watchdog design (follow-on work to the twin incidents of 2026-07-07 and 2026-08-18, BOB-116 / BOB-120 in the SSoT), one of the four architectural options I enumerated was:

Option 4 “ rootless podman container. Run the watchdog inside a rootless podman container so it lives OUTSIDE the user@1000.service scope and SURVIVES a SIGKILL cascade of that unit. Attractive because it reuses the boba container substrate (Â§11.4.161) and needs no privileged systemd unit.

The load-bearing claim was “lives OUTSIDE user@1000.service and SURVIVES the SIGKILL cascade.” That claim was WRONG.

2. Why it was wrong “ captured evidence

Rootless podman on this host runs **inside** the user@1000.service cgroup, not outside it. Every rootless container’s cgroup path is a child of / user.slice/user-1000.slice/user@1000.service/. When user@1000.service receives SIGKILL, systemd walks the whole cgroup subtree and SIGKILLS every descendant “ the container’s common, its init, and every process inside it “ in the same cascade that takes down gnome-shell, ssh, gvfsd, and every other user-session process.

Captured on this host 2026-08-19T11:34Z (this very session):

```
$ podman info | grep -iE 'rootless|cgroup'
cgroupControllers:
cgroupManager: systemd
cgroupVersion: v2
rootlessNetworkCmd: pasta
rootless: true
```

```
$ systemctl show user@1000.service | grep -E '^(Id|Slice)='
Id=user@1000.service
Slice=user-1000.slice
```

```
$ podman inspect helix-nats | grep CgroupPath
"CgroupPath": "/user.slice/user-1000.slice/user@1000.service/user.slice/li
$ podman inspect helix-redis | grep CgroupPath
"CgroupPath": "/user.slice/user-1000.slice/user@1000.service/user.slice/li
$ podman inspect helix-postgres | grep CgroupPath
"CgroupPath": "/user.slice/user-1000.slice/user@1000.service/user.slice/li
```

Every rootless container running on this host is a descendant of user@1000.service in cgroup v2. A SIGKILL delivered to user@1000.service propagates down the cgroup tree to each of them.

Corroborating physical evidence from the 3rd forced-logout (BOB-120, 2026-08-18 23:45:49): the 73-line Killing process cascade captured in docs/qa/BOB-120/journalctl_23-42_to_23-46.log explicitly names the boba containers jackett, qbittorrent-nox, and the boba-jackett Python worker (PID 3111634) plus its common (PID 2319406) "proof that live rootless containers on THIS host DID die together with user@1000.service in the observed real event, not merely in theory. A watchdog placed inside such a container would have died with them.

3. Rule violated

Â§11.4.6 **no-guessing** applied to architecture-analysis: "rootless container survives the kill" was an ARCHITECTURAL claim, not a captured-evidence statement. It was carried forward from a general intuition that "rootless" user-scoped "not privileged" somehow separate from the user session without ever being verified against the cgroup hierarchy the local host actually publishes. The word "SURVIVES" belongs to Â§11.4.6's forbidden vocabulary until proven "never" "presumably survives", never "should survive", never asserted without podman inspect | grep CgroupPath and a systemctl show cross-check on the SPECIFIC host being reasoned about. It was, in the anchor's own terms, exactly the class of guess Â§11.4.6 forbids.

Â§11.4.194 **exhaustive review** was also violated: an exhaustive all-angles review would have asked "what cgroup slice does rootless podman put a container in on THIS host?" as an explicit precondition of the SURVIVES claim, and refused GO until that condition was proven from podman inspect output. Instead the option list was drafted assuming an adjacent-but-

different fact (rootful containers under machine.slice survive) applied to rootless – a §11.4.112(5)-class adjacent-goal leak of a bounded verdict beyond its evidence fence.

§11.4.226 **evidence-class-at-closure** completes the picture: a runtime/hardware claim (‘‘this container survives when this cgroup is killed’’) can never close on source-class or intuition-class evidence – only on runtime evidence read from the actual host’s cgroup topology.

4. Correct architecture

Options that GENUINELY sit outside user@1000.service and survive its SIGKILL cascade – validated on this host by inspecting where their PIDs actually reside:

- **Option A – root systemd unit.** A .service under /etc/systemd/system/ runs under system.slice, not under user-1000.slice. Requires sudo once to install; requires an ExecStart= binary that itself needs no interactive credentials (the operator’s canonical constraint).
- **Option D – separate UID.** A dedicated non-login system user (e.g. boba-watchdog) with its own user@<uid>.service scope. When user@1000.service dies, user@<other-uid>.service is untouched.

Both are structurally-separated in the cgroup sense the original Option 4 claim needed and did not have. Option-C rootful-podman would also qualify (containers land under machine.slice), but is ruled out independently by §11.4.161 rootless-container-runtime mandate.

The Option-4 rootless-container proposal is WITHDRAWN and MUST NOT be re-proposed for this watchdog without new host-specific evidence that the host’s cgroup topology has fundamentally changed (e.g. a future podman/systemd release moving rootless containers to a non-user slice, proven by fresh podman inspect | grep CgroupPath output – never assumed).

5. Preventive mechanism – how §11.4.194 + §11.4.6 SHOULD have caught this

The failure was a MISSING VERIFICATION STEP between ‘‘propose an architectural option’’ and ‘‘commit the option list to a decision doc.’’ Concretely:

1. **Every architectural option that claims survival, isolation, or scope-separation from a named process/scope MUST cite a captured cgroup/slice/namespace inspection of THAT scope on the target host.** No such citation – the option is UNCONFIRMED per §11.4.6 and MUST be labelled so in the doc, never listed alongside verified options as if peer-verified.

2. **The Â§11.4.194 exhaustive-review checklist for architecture docs MUST include an explicit “prove-the-boundary” line item:** for every claim of the form “X sits outside Y and survives when Y dies”, the review captures (a) X’s current cgroup / namespace / slice on the target host, (b) Y’s cgroup / namespace / slice, (c) a proof from (a)+(b) that X is NOT a descendant of Y in every relevant hierarchy (cgroup, PID namespace, mount namespace when applicable).
3. **Â§11.4.201 golden-FALSE fixture for architecture docs:** an architectural option list should be reviewable by a mutation that INJECTS a plausible-sounding-but-wrong SURVIVES claim (exactly the Option-4 shape) and asserts the review FAILs it. If the review would have let the mutation through “as it did here” the review process itself is the defect, not just this one option.
4. **Recording it here so the next architecture doc inherits the correction** “Â§11.4.138’s permanent-guard-per-escape discipline applied to architecture-analysis: this doc IS the guard, cited from the forced-logout preventive-watchdog design doc when it is next edited, so a future author looking at “why is Option 4 gone?” sees the refutation and the captured evidence, not silence.

6. What did not change

- The forced-logout incident record itself (BOB-116, BOB-120) is unaffected “this correction concerns a proposed RESPONSE architecture, not the incident findings.
- The Â§11.4.161 rootless-container mandate is unaffected “rootless podman remains the correct runtime for every OTHER boba container workload; the rejection is specific to “put a survives-user-session-death watchdog inside one”.
- No production code shipped with the bluff; it was caught at the architecture-proposal stage.

Sources verified 2026-08-19 (per Â§11.4.99):

- `podman info on host` MilosVasic “rootless: true, cgroupManager: systemd, cgroupVersion: v2
- `systemctl show user@1000.service` “Slice=user-1000.slice
- `podman inspect helix-nats|helix-redis|helix-postgres` “each CgroupPath a descendant of /user.slice/user-1000.slice/user@1000.service/”
- `docs/qa/BOB-120/journalctl_23-42_to_23-46.log` “real-event observation of jackett / qbittorrent-nox / boba-jackett python + common dying in the user@1000.service SIGKILL cascade of 2026-08-18 23:45:49
- `docs/incidents/2026-08-18-3rd-forced-logout.md` “the incident record cross-referencing the containers-in-kill-list finding
- Constitution Â§11.4.6, Â§11.4.138, Â§11.4.161, Â§11.4.194, Â§11.4.201, Â§11.4.226, Â§11.4.112(5) “cited above